How the Etter Value Model works

Every step, formula and assumption, left to right. Moderate, Attrition only, cross-industry, Etter FP&A data, agent ceilings 90% / 80% (Chandan, 23 Oct); updated 5 November 2026 (NEW = changed, see page 2). Rust = cash (fewer positions). Green = capacity (time back). Blue = hires avoided. Grey = valued at zero. A line in the box colour marks a formula.

1 WHAT GOES IN

Etter Tasks sheet

420 tasks across 11 roles. For each task: the **role**, its **workload** (a group of related tasks), its **category** (one of six) and the **hours one person spends on it a month**.

Why this sheet: every task has its own hours, category and description, so every number traces back to a named task. Etter's Workloads sheet splits the same hours differently across categories; still an open question for Etter.

Etter Roles sheet + client

For each role: **headcount** (1,523 in total), **average base salary** and **level**, which maps to a turnover band.

Etter's own automation score for each role is compared with the task hours as a second opinion (flagged if more than 10 points apart), never mixed into the numbers.

Four choices, then the dials

The client picks a **scenario**, a **policy** for freed time (box 5), an **industry**, and whether to include **assistant seats**. Every other number sits on Inputs with a source and a grade: A official figure, B benchmark, C estimate, D thin evidence.

Uncertain dials switch together by scenario:

Dial	Cons.	Mod.	Aggr.
AI takeover	25%	40%	60%
Human review	40%	30%	20%
Review rise per band NEW	20%	20%	20%
Months to full use	18	12	6
Assistant active use	55%	70%	85%
Cost multiplier	1.25	1.00	0.90

Industry packs change only the discount rate, staff turnover and job growth (healthcare 6.19%, ×0.91, 0.9%; insurance 6.34%, ×0.59; tech 9.34%, ×0.59; default 6.96%, ×1.00, 0%). Company size enters as headcount, not as a setting.

2 COST OF AN HOUR, KIND OF WORK

Loaded hourly rate

> hourly rate = salary × 1.31 ÷ (12 × task hours a month)

1.31: of every $100 an employer spends on a full-time private-industry worker, $68.50 is wages and $8.10 paid leave (BLS ECEC, Table 5). A salary already pays for leave days, so 100 ÷ (68.5 + 8.1) = 1.31.

160 hours a month (1,920 a year), the Etter file's convention. Senior Analysts' tasks cover 136; shares are taken of the hours recorded.

Pay grows 3.4% a year (BLS Employment Cost Index) NEW: salaries saved, capacity, severance and training time follow it. The cost test stays at today's prices.

Example, Analysts: $79,661 × 1.31 = $103,997 a year ÷ 1,920 = $54.17 an hour.

Sort by category

Category	Most AI could do	Kind
Directive	**90%**	Agent
Feedback Loop	**80%**	Agent
Task Iteration	45%	Assistant
Learning	25%	Assistant
Validation	50%	Assistant
Negligibility	0%	Human

An **agent** does the work instead of the person, so its time can become fewer positions. An **assistant** makes the person faster, so its time is spread thin and stays as capacity by default.

By job band

Band	Assistant saving	Review NEW	Leaving / yr
Para-professional	20%	24%	12.5%
Professional	12%	30%	9.1%
Management	6%	36%	6.3%
Executive	3%	42%	5.2%

Assistants help juniors most (Brynjolfsson, Li and Raymond, QJE 2025). Senior work needs more checking (Brynjolfsson; Dell'Acqua 2026): review is the measured 30% at Professional level, ±20% of it per band (the size is judgment, grade D). Leaving rates: Mercer 2025, × the industry's turnover factor (BLS JOLTS quits).

3 WHAT AI COULD DO, WHAT TO BUILD

Agent work (Directive, Feedback Loop)

> AI-able hours = hours a month × headcount × most AI could do

Example, Analysts' Extract and format financial data: 8.64 × 310 × 90% = 2,411 hours a month (28,927 a year).

Is the AI cheaper than the work?

One agent per workload (59 have agent tasks), shared by every role that does it, priced two ways:

Build: $55,000, or $130,000 if 2+ roles or 3+ agent tasks; $3.48 per AI hour; 18% a year upkeep. **Buy:** $15,000 / $35,000; $6 per AI hour.

> value a month = AI-able hours × hourly rate
> NPV at cost = (value × each role's net realisation − running − upkeep) × 42.5 − up front

The cheaper option is kept; the workload passes if its NPV is above zero. **42.5:** months of full value in five years after the 3-month build, the 12-month ramp-up and 6.96% discounting, at today's pay and prices.

Financial analysis and reporting: 5 roles, $1,016,529 of work a month; build $10.69M vs buy $10.02M, so build. Cost review planning and support: $3,850 a month; buy, +$15,884, passes but ranks last.

How big a programme

Passing workloads are ranked by return per dollar up front and added one at a time. The programme stops where **five-year cash peaks: 35 of 59** (25 bought, 10 built, $1.14M up front).

Cash counts only whole positions, so late, small workloads add cost but free fractions of people that never round up.

Go-live: pilot of the top 3 in month 3, a 6-month evaluation, then waves of 10 in months 12, 15, 18 and 21. **Gate, month 9** NEW: scale only if the pilot reaches the 19.2% break-even; otherwise stop.

Assistant work (Task Iteration, Learning, Validation)

> hours saved = hours × smaller of (most AI could do, band saving) × 70% active use

Seats roll out from month 2 over 6 months, $360 a seat a year for people in post, trued down each contract year NEW. Example, Analysts' Develop custom financial analyses: 21,427 × 12% × 70% = 1,800 hours.

Stays human (Negligibility): no saving

4 WHAT AI REALLY FREES

AI takes over, people check it

> hours freed = AI-able hours × takeover × (1 − review)

Takeover 40%: the share of AI-able work the agent actually ends up doing (range 20–62%, grade C). **Review 30%:** for every 100 hours it does, a Professional spends 30 checking (15–45%, C); by band 24% / 30% / 36% / 42% NEW.

40% × 70% = **28% net realisation**, the one number the answer turns on. Running cost is paid on all hours the AI works, reviewed or not.

Example: 28,927 × 40% = 11,571 AI hours; × 70% = 8,099 hours freed a year.

Ramped in, wave by wave

> use in month m = S-curve from go-live, full after 12 months

The same curve drives the cost test, the running cost and the savings (MIT NANDA 2025: 6 to 24 months).

The work left behind is harder

> freed × (1 − 0.30 × share of the role's time freed)

Complexity drag, applied to agent and assistant time together. Grade D: a simple stand-in for the exception tail until Etter gives volumes or handling times.

Analysts free about 14% of their time, so they keep 96%: 8,099 becomes 7,769 hours, 4.0 people's worth.

Why AI-able hours are not headcount

All roles, month 60	People's worth
Hours in agent categories	563.7
× most AI could do	482.0
In the 35 workloads	452.3
× 40% takeover	180.9
after review (by band)	124.6
After complexity drag	119.9
Positions not refilled	**56**

Plus 52.3 people's worth from assistants. 172 freed in all.

5 WHERE EACH FREED HOUR GOES

Policy: the company's choice

	Capacity only	Attrition only	Active reduction
Agent time to positions	0%	**50%**	75%
Assistant time	0%	**0%**	25%
How	nobody goes	as people leave	2 months' notice, 12 weeks' severance

The harvest share moves cash more than any other input (−$4.8M at 25% to +$18.4M at 75%). Healthcare defaults to capacity only.

Cash: positions not refilled

> target = (harvest × freed people), rounded down, per role

> people who have left by month m = headcount × (1 − (1 − yearly rate)$^{m \div 12}$)

> positions in month m = smaller of (target × progress so far) and (people who have left)

A removed position stays removed while the AI runs and never exceeds the time freed; if the programme stops, the work comes back and each is refilled at $4,700 NEW. A role needs roughly 20–30 people before agents free a whole position.

Example, Analysts: 28.1 freed by agents × 50% = 14.0, so 14. 118 Analysts leave in five years: "leave 14 of the next 118 leavers unreplaced". All roles: 56 positions, 3.7% of FP&A, by month 33.

Hires avoided: against the plan

> avoided = smaller of (spare whole people, planned hires so far)

Only if the industry is growing: spare freed time fills planned hires before it becomes capacity. FP&A has 0% growth, so 0; the healthcare pack avoids 63.

Capacity: time back NEW

> left over = freed − (positions + hires avoided) × hours

> capacity = left over − 8 hours a month × people in post

At the hourly rate, growing with pay: 45 people's worth at month 60. Every month, positions + hires avoided + capacity + stranded = hours freed, to 0.00000 hours.

Stranded: too thin to use NEW

Every role frees only 4.5–14.8 hours a month per person still in post. The first 8 (about 24 minutes a day) is slack nobody can redeploy: 72 people's worth, valued at zero.

6 MONEY, ODDS AND THE DECISION

Costs, five-year present value

Training: $846 + 16.7 hours each: 1,523 at first use, 581 joiners, 25% refresher a year NEW	$6.53M
Running: $3.48 / $6 per AI hour, 13% cheaper a year	$2.64M
Assistant licences: $360 a seat a year, people in post	$1.45M
Build or buy, spread over the 3-month lead	$1.06M
Programme: $75,000 + $623 per person (Forrester)	$1.02M
Upkeep of custom agents: 18% a year	$0.41M
Total (severance only under active reduction)	**$13.12M**

Three kinds of money, never added

> each month's money × 1.0696$^{-m \div 12}$, added up over 60 months

Salaries no longer paid (× 1.31, +3.4% a year)	$19.83M
Recruiting avoided ($4,700 per leaver not replaced) + seats	$0.31M
minus costs	$13.12M
Cash vs today, NPV	**$7.02M**
Hires avoided vs plan	$0
Capacity at cost	$14.5M

Payback month 38 discounted, 36 undiscounted NEW; peak funding $4.7M. If nobody leaves, cash is −$13.83M against $33.6M of capacity: **41%** of that time must become real value to break even.

How sure: 1,000 futures

Each run draws all twelve uncertain inputs from their ranges; good programmes are good at everything at once (correlation 0.5). Fixed seed, so the same answer every time.

60% chance cash is positive · P10 −$4.8M · P90 +$20.7M

Break-even: **19.2%** net realisation (we assume 28%). **28%** of futures miss it at the month-9 gate and stop NEW: AI off, work back, −$4.4M.

The decision

Go at 70% or better, pilot first at 40–70%, stop below 40%.

PILOT FIRST the top 3 go live in month 3 and are measured for six months. Scale only if net realisation reaches 19.2%; otherwise stop. Watch takeover, checking time by level, adoption speed and real costs.

Conservative −$4.1M (17%, stop) · Aggressive +$25.6M (95%) · Active reduction +$21.8M (92%).

Checks: twenty-one tie-outs to the Etter file and self-checks, all pass; hours reconcile to 0.00000 every month; the Excel matches the engine to the cent ($7,023,654.78). Sources: Etter FP&A task file; BLS ECEC, ECI, JOLTS; Brynjolfsson et al. 2025; Dell'Acqua et al. 2026; Mercer 2025; MIT NANDA 2025; Damodaran 2026; Microsoft; Forrester; ATD 2026; BCG 2025; Gartner 2024; SHRM.

What changed since 23 October

For the review with Chandan. Same data and defaults as page 1. Each change was written down as a spec before it was coded (docs/Etter_Value_Model_spec_delta_review.md), is a live formula in the Excel, and matches the engine to the cent. The 23 October model can still be reproduced exactly (legacy switch).

A THE HEADLINE, STEP BY STEP

From $9.38M to $7.02M

Five-year cash NPV, $ thousands	Change	Cash
23 October model (Go, 75% chance positive)		9,376
Train joiners as they arrive	−935	
Licences follow people in post	+36	
Yearly training refresher, 25%	−2,331	6,146
Capacity floor, stop path, pilot gate	none	
Wage growth, 3.4% a year	+1,927	8,073
Review rises with seniority	−1,049	
Now (PILOT FIRST, 60% chance positive)		**7,024**

The programme stays at 45 workloads until the last step; review by seniority makes management-heavy workloads free less, and the peak moves to 35.

Before and now

	23 Oct	Now
Cash vs today, 5-year NPV	**$9.38M**	**$7.02M**
Positions not refilled	59	56
Workloads (bought)	45 (35)	35 (25)
Capacity at cost	$47.7M	$14.5M
Payback month, discounted / undiscounted	34 / 33	38 / 36
Chance cash is positive	75%	60%
Break-even net realisation	14.3%	19.2%
Verdict	Go	PILOT FIRST

Training the joiners alone takes the chance below the 70% line for Go. That fix is structural, not a judgment call.

B FIVE FIXES FROM THE FINANCE REVIEW

1 Capacity. Positions were whole people, but every leftover sliver counted as capacity at full cost. Now freed time below 8 hours a month per person is stranded at zero value.
Capacity $14.5M with the floor, $50.7M without it; cash unchanged.

2 Training. It was paid once, for the day-one headcount. Now 581 joiners are trained as they arrive, and everyone in post gets a 25% refresher each year.
−$0.93M joiners, −$2.33M refresher.

3 Licences. Seats stayed sized on the day-one headcount of 1,523 for all five years. Now they follow people in post and are trued down at each contract anniversary.
+$0.04M.

4 Payback. It was undiscounted next to a discounted NPV. Now both are shown and labelled.
Month 38 discounted, 36 undiscounted.

5 Stopping. The model could not show a programme that was tried and stopped. Now it can: AI off, the work comes back, positions are refilled at $4,700 each. A gate at month 9 stops every future that misses the break-even.
28% of futures stop, −$4.4M each. Gartner forecast at least 30% of GenAI projects abandoned after proof of concept.

C TWO PARAMETERS ADDED

Wage growth, 3.4% a year (BLS Employment Cost Index, sourced). Salaries saved, capacity, severance and training time grow with pay; the cost test stays at today's prices.
+$1.93M. Range 2.5–4.5%: +$6.5M to +$7.6M (9th of 13 in the tornado).

Review rises with seniority, 20% a band (judgment; direction from Brynjolfsson v2 and Dell'Acqua 2026). Review is 24% / 30% / 36% / 42% from Para-professional to Executive.
−$1.05M; programme 45 → 35 workloads, positions 59 → 56. Range 5–40%: +$7.0M to +$5.5M (8th of 13).

Both pass the working agreement's test for a new parameter: a top-ten mover in the tornado. At zero, each gives back the model before it exactly.

Not changed

The six steps; the Etter data; ceilings 90% / 80%; the three policies; Go at 70%, pilot first at 40–70%; the top three movers (harvest share, takeover, review); cash, hires avoided and capacity never added together.

D FOR CHANDAN TO DECIDE

Questions only Chandan can answer

1. **Ceilings.** The source workbook states the Etter ceilings twice, and the tables disagree on five of six categories. What does Feedback Loop at 80% rest on? Cash under each reading: Tasks column $3.44M · in use $7.02M · Read me table $7.23M · bucket $10.65M.
2. **Tax.** Pre-tax or after-tax cash flows? The discount rate is post-tax; the cash flows are pre-tax.
3. **Month 61.** Does the programme carry any residual value after five years? Today it stops at month 60.

Three judgment calls to confirm

1. **Smallest usable block of freed time: 8 hours a month per person.** Capacity is $50.7M at 0 and $0 at 16 hours. Cash does not move.
2. **Training refresher: 25% of the initial package a year.** Cash +$8.5M at 10%, +$4.5M at 50%.
3. **Review rise: 20% a band.** Cash +$7.0M at 5%, +$5.5M at 40%. The pilot can replace it: measure checking time by level.

Engine, Excel and website agree on every number here (38 tests, parity 1.3×10^{-14}, Excel to the cent). The 23 October chance of positive cash and verdict are quoted from that run; its cash, positions, workloads, capacity, paybacks and break-even are recomputed from the legacy switch.